IGCSE Chemistry CIE

YOUR NOTES

2. Atoms, Elements & Compounds

CONTENTS

2.1 Atomic Structure & the Periodic Table

2.1.1 Elements, Compounds & Mixtures

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Elements, Compounds & Mixtures

Elements, compounds and mixtures

- All substances can be classified into one of these three types

Element

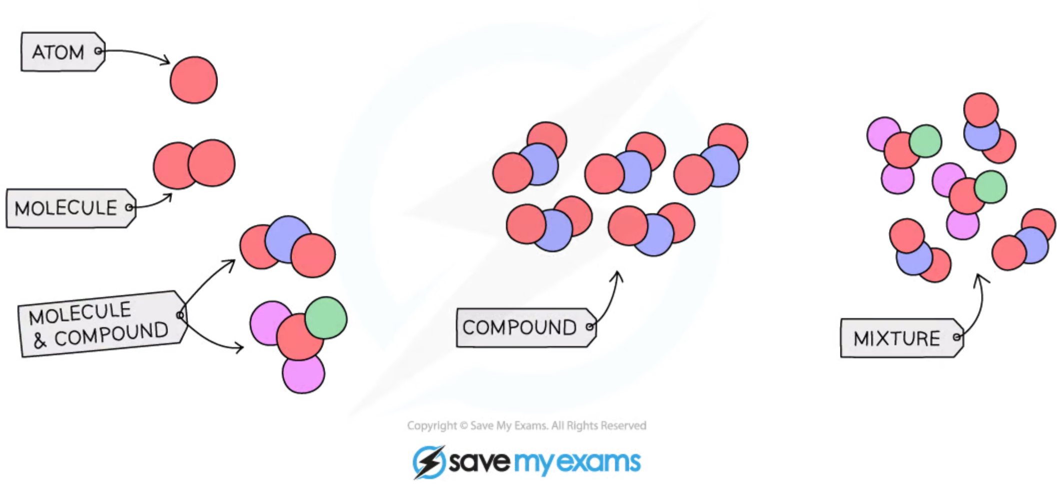
- A substance made of atoms that all contain the **same number of protons** and cannot be split into anything simpler
- There are 118 elements found in the Periodic Table

Compound

- A pure substance made up of two or more elements **chemically combined**
- There is an **unlimited** number of compounds
- Compounds cannot be separated into their elements by physical means
- E.g. copper(II) sulfate ($CuSO_4$), calcium carbonate ($CaCO_3$), carbon dioxide (CO_2)

Mixture

- A combination of two or more substances (elements and/or compounds) that are **not** chemically combined
- Mixtures can be separated by **physical methods** such as filtration or evaporation
- E.g. sand and water, oil and water, sulfur powder and iron filings



Particle diagram showing elements, compounds and mixtures

2.1.2 Atomic Structure

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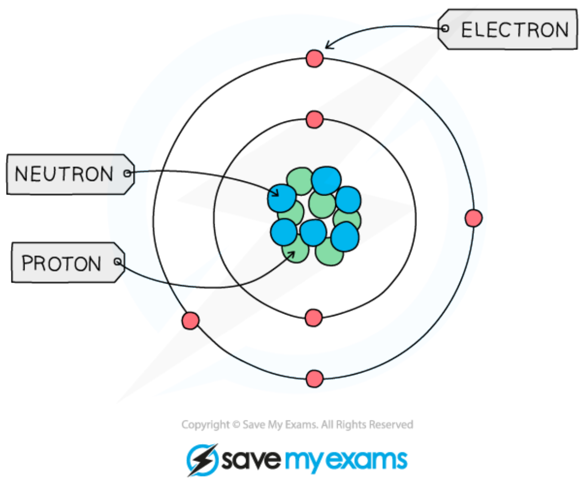
Atomic Structure

- All substances are made of tiny particles of matter called **atoms** which are the building blocks of all matter
- Each atom is made of subatomic particles called **protons**, **neutrons,** and **electrons**
- The protons and neutrons are located at the centre of the atom, which is called the **nucleus**
- The electrons move very fast around the nucleus in orbital paths called **shells**
- The mass of the electron is negligible, hence the mass of an atom is contained within the nucleus where the protons and neutrons are located



The structure of the carbon atom

Protons, Neutrons & Electrons

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- The size of atoms is so tiny that we can't really compare their masses in conventional units such as kilograms or grams, so a unit called the relative atomic mass is used
- One **relative atomic mass** unit is equal to 1/12th the mass of a carbon-12 atom.
- All other elements are measured relative to the mass of a carbon-12 atom, so relative atomic mass has no units
- Hydrogen for example has a relative atomic mass of 1, meaning that 12 atoms of hydrogen would have exactly the same mass as 1 atom of carbon
- The relative mass and charge of the sub-atomic particles are shown below:

Table of Subatomic Particles

Particle	Relative mass	Charge
Proton	1	1+
Neutron	1	0 (Neutral)
Electron	$\frac{1}{1840}$	1–

Exam Tip

Knowing the exact mass of an electron is not in the specification and saying it is almost nothing or negligible will be sufficient. It does, however, sometimes appear in particle identification questions, but you can usually deduce that it is the electrons from other information in the question.

Defining Proton Number

- The atomic number (or **proton number**) is the number of protons in the nucleus of an atom
- The symbol for atomic number is ***Z***
- It is also the number of electrons present in a neutral atom and determines the **position** of the element on the Periodic Table

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Defining Mass Number

- The Nucleon number (or **mass number**) is the total number of protons **and** neutrons in the nucleus of an atom
- The symbol for nucleon number is ***A***
- The nucleon number **minus** the proton number gives you the number of **neutrons** of an atom
- Note that protons and neutrons can collectively be called **nucleons**.
- The atomic number and mass number of an element can be shown using **atomic notation**
- The **Periodic Table** shows the elements together with their atomic (proton) number at the top and relative atomic mass at the bottom – there is a difference between relative atomic mass and mass number, but for your exam, you can use the relative atomic mass as the mass number (with the exception of chlorine)

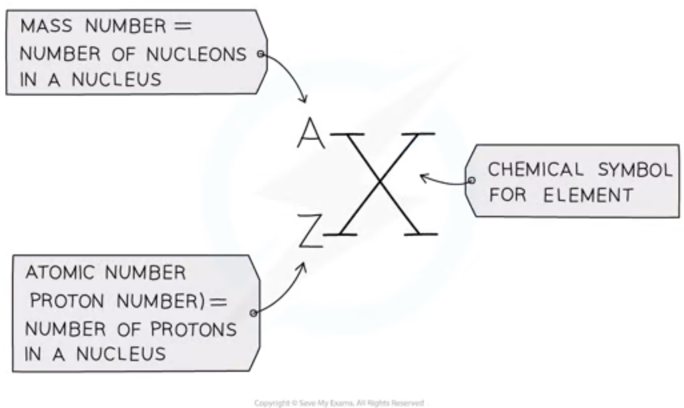
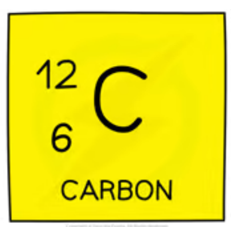


Diagram showing atomic notation



Atomic notation for carbon

Exam Tip

Both the atomic number and the relative atomic number (which you can use as the mass number) are given on the Periodic Table but it can be easy to confuse them. Think **MASS** = **MASSIVE**, as the mass number is **always** the bigger of the two numbers, the other smaller one is thus the atomic / proton number. **Beware** that some Periodic Tables show the numbers the other way round with the atomic number at the bottom!

Deducing protons, neutrons & electrons

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Finding the protons

- The atomic number of an atom and ion determines which element it is
- Therefore, all atoms and ions of the **same element** have the same number of protons (atomic number) in the nucleus
  - E.g. lithium has an atomic number of 3 (three protons) whereas beryllium has atomic number of 4 (4 protons)

- The number of protons equals the **atomic (proton) number**
- The number of protons of an **unknown** element can be calculated by using its mass number and number of neutrons:

Mass number = number of protons + number of neutrons

Number of protons = mass number – number of neutrons

Finding the electrons

- An atom is **neutral** and therefore has the **same** number of **protons** and **electrons**

Finding the neutrons

- The **mass** and **atomic numbers** can be used to find the number of **neutrons** in **ions** and **atoms:**

Number of neutrons = mass number – number of protons

Worked Example

Determine the number of protons, electrons and neutrons in an atom of element X with atomic number 29 and mass number 63

Answer:

- The number of protons of element **X** is the same as the atomic number

Number of protons = 29

- The **neutral atom** of element **X** therefore also has **29 electrons**
- The atomic number of an element **X** atom is 29 and its mass number is 63

Number of neutrons = mass number – number of protons

Number of neutrons = 63 – 29

Number of neutrons = 34

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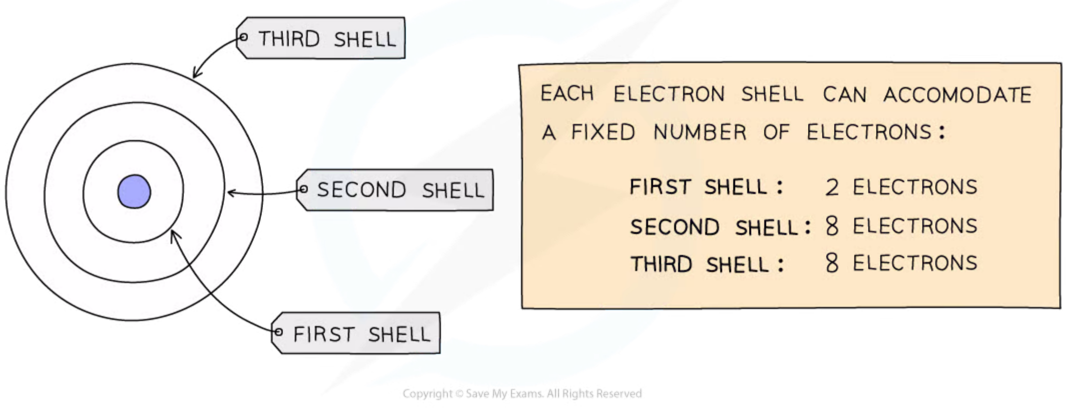
Electronic Configuration

Electronic configuration

- We can represent the structure of the atom in two ways: using diagrams called **electron shell diagrams** or by writing out a special notation called the **electronic configuration** (or electronic structure or electron distribution)

Electron shell diagrams

- Electrons orbit the nucleus in **shells** (or **energy levels**) and each shell has a different amount of energy associated with it
- The further away from the nucleus, the more energy a shell has
- Electrons fill the shell closest to the nucleus
- When a shell becomes full of electrons, additional electrons have to be added to the next shell
- The first shell can hold 2 electrons
- The second shell can hold 8 electrons
- For this course, a simplified model is used that suggests that the third shell can hold 8 electrons
    - For the first 20 elements, once the third shell has 8 electrons, the fourth shell begins to fill
- The outermost shell of an atom is called the **valence** shell and an atom is much more stable if it can manage to completely fill this shell with electrons



A simplified model showing the electron shells

- The arrangement of electrons in shells can also be explained using numbers
- Instead of drawing electron shell diagrams, the number of electrons in each electron shell can be written down, separated by commas
- This notation is called the electronic configuration (or electronic structure)
    - E.g. Carbon has 6 electrons, 2 in the 1st shell and 4 in the 2nd shell
        - Its electronic configuration is 2,4

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- Electronic configurations can also be written for ions
  - E.g. A sodium atom has 11 electrons, a sodium ion has lost one electron, therefore has 10 electrons; 2 in the first shell and 8 in the 2nd shell
    - Its electronic configuration is 2,8

The Electronic Configuration of the First Twenty Elements

ELEMENT	ATOMIC NUMBER	ELECTRONIC CONFIGURATION
HYDROGEN	1	1
HELIUM	2	2
LITHIUM	3	2,1
BERYLLIUM	4	2,2
BORON	5	2,3
CARBON	6	2,4
NITROGEN	7	2,5
OXYGEN	8	2,6
FLUORINE	9	2,7
NEON	10	2,8
SODIUM	11	2,8,1
MAGNESIUM	12	2,8,2
ALUMINIUM	13	2,8,3
SILICON	14	2,8,4
PHOSPHORUS	15	2,8,5
SULFUR	16	2,8,6
CHLORINE	17	2,8,7
ARGON	18	2,8,8
POTASSIUM	19	2,8,8,1
CALCIUM	20	2,8,8,2

Note: although the third shell can hold up to 18 electrons, the filling of the shells follows a more complicated pattern after potassium and calcium. For these two elements, the third shell holds 8 and the remaining electrons (for reasons of stability) occupy the fourth shell first before filling the third shell.

Exam Tip

You need to be able to write the electronic configuration of the first twenty elements and their ions. You may see electronic configurations using full stops or '+' signs instead of commas. You would not be penalised for using full stops.

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Electron Shells & The Periodic Table

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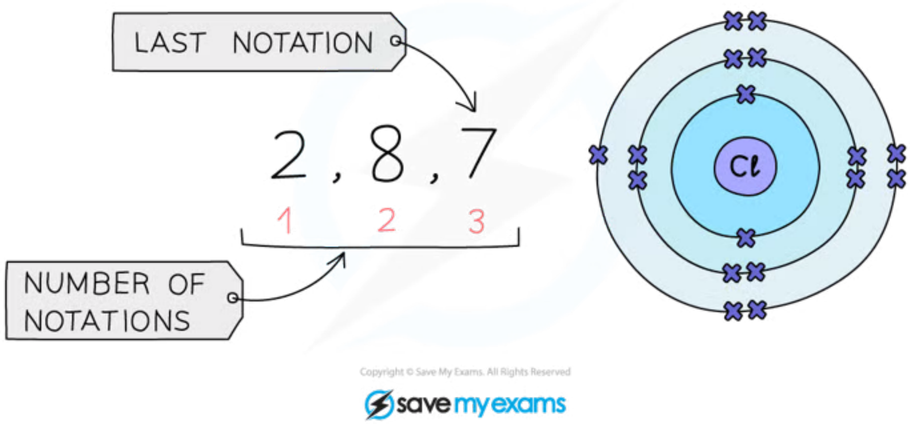
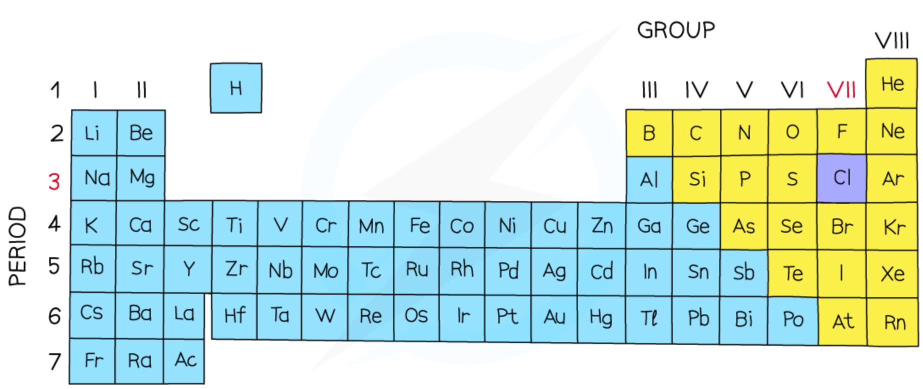
- There is a clear **relationship** between the electronic configuration and how the Periodic Table is designed
- The number of notations in the electronic configuration will show the number of occupied shells of electrons the atom has, showing the **period** in which that element is in
- The last notation shows the number of outer electrons the atom has, showing the **group** that element is in (for elements in Groups I to VII)
- Elements in the **same group** have the **same** number of outer **shell electrons**



The electronic configuration for chlorine

Period: The red numbers at the bottom show the number of notations which is 3, showing that a chlorine atom has 3 occupied shells of electrons and is in Period 3

Group: The final notation, which is 7 in the example, shows that a chlorine atom has 7 outer electrons and is in Group VII



The position of chlorine on the Periodic Table

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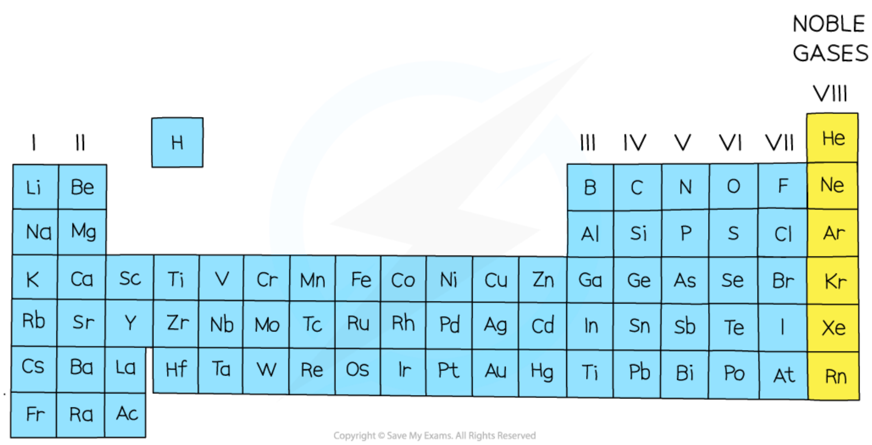
- In most atoms, the outermost shell is not full and therefore these atoms react with other atoms in order to achieve a full outer shell of electrons (which would make them more stable)
- In some cases, atoms lose electrons to entirely empty this shell so that the next shell below becomes a (full) outer shell
- All elements wish to fill their outer shells with electrons as this is a much more stable configuration

The noble gases

- The atoms of the Group VIII elements (the noble gases) all have a full outer shell of electrons
- All of the noble gases are **unreactive** as they have full outer shells and are thus very stable



The noble gases are on the Periodic Table in Group 8/0

Exam Tip

The electrons in the outer shell are also known as valency electrons.

2.1.4 Isotopes

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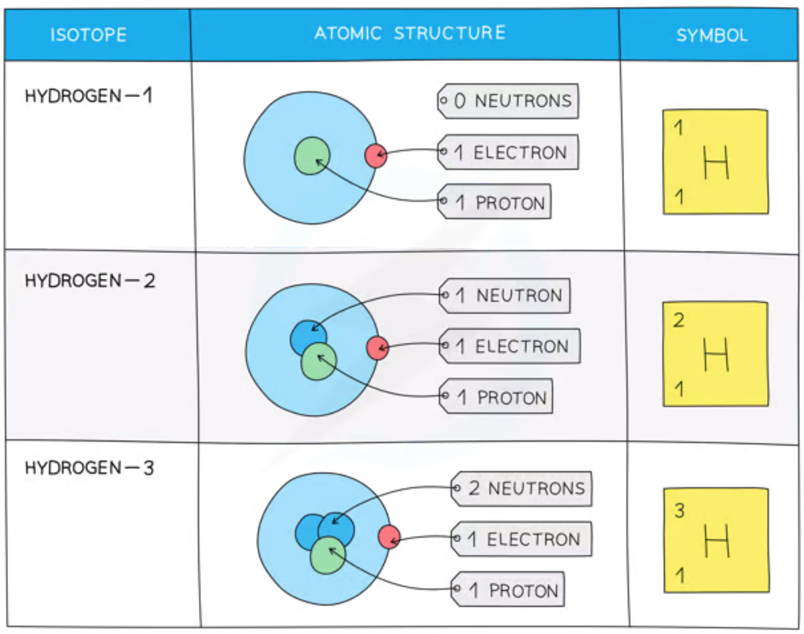
Defining Isotopes

- Isotopes are different atoms of the **same element** that contain the same number of **protons** but a different number of **neutrons**
- The symbol for an isotope is the chemical symbol (or word) followed by a dash and then the mass number
- So C-14 (or carbon-14) is the isotope of carbon which contains 6 protons, 6 electrons and 14 – 6 = 8 neutrons
  - It can also be written as ^{14}C or $^{14}_{6}C$

The Atomic Structure and Symbols of the Three Isotopes of Hydrogen



Why Isotopes Share Properties

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- Isotopes of the same element display the **same chemical characteristics**
- This is because they have the same number of electrons in their outer shells and, therefore, the same electronic configuration and this is what determines an atom's chemistry
- The difference between isotopes is the number of neutrons which are neutral particles within the nucleus and add mass only
- The difference in mass affects the physical properties, such as density, boiling point and melting point
- Isotopes are identical in appearance, so a sample of C-14 would look no different from C-12
- Water made from deuterium oxide is known as 'heavy' water, and has a relative formula of mass 20, compared to 18 for water, so it is 20% heavier, but it would look, taste and feel just like normal water
    - However, it wouldn't be a good idea to drink it because it is toxic as it interferes with biochemical reactions in your cells!

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Calculating Relative Atomic Mass

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Relative Atomic Mass

- The symbol for the relative atomic mass is A_r
- The relative atomic mass for each element can be found in the Periodic Table along with the atomic number
- The atomic number is shown above the atomic symbol and the relative atomic mass is shown below the atomic symbol
- Atoms are too small to accurately weigh but scientists needed a way to compare the masses of atoms
- The carbon-12 is used as the standard atom and has a fixed mass of 12 units
- It is against this atom which the masses of all other atoms are compared
- Relative atomic mass (A_r) can therefore be defined as:
  - **the average mass of the isotopes of an element compared to 1/12th of the mass of an atom of ^{12}C**

- The relative atomic mass of carbon is 12
  - The relative atomic mass of magnesium is 24 which means that magnesium is twice as heavy as carbon
  - The relative atomic mass of hydrogen is 1 which means it has one-twelfth the mass of one carbon-12 atom
- The relative atomic mass of an element can be calculated from the **mass number** and **relative abundances of all the isotopes** of a particular element using the following equation:

$$A_r = \frac{(\%\text{ of isotope 1 x mass number of isotope 1}) + (\%\text{ of isotope 2 x mass number of isotope 2})}{100}$$

- The top line of the equation can be extended to include the number of different isotopes of a particular element present.

Example

The table shows information about the isotopes in a sample of rubidium

ISOTOPE	NUMBER OF PROTONS	NUMBER OF NEUTRONS	PERCENTAGE OF ISOTOPE IN SAMPLE
1	37	48	72
2	37	50	28

$$A_r = \frac{(72 \text{ x } 85) + (28 \text{ x } 87)}{100} = 85.6$$

Is mass number and relative atomic mass the same thing?

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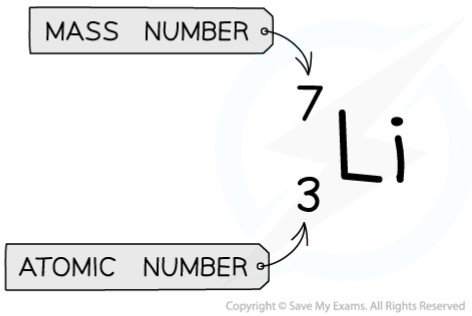
- On the Periodic Table provided in your exam you will see that lithium has a relative atomic mass of **7**
- Although it seems that this is the same as the mass number, they are not the same thing because the relative atomic mass is a **rounded** number
- Relative atomic mass takes into account the existence of isotopes when calculating the mass
- Relative atomic mass is an **average mass** of all the isotopes of that element
- For simplicity **relative atomic masses** are often shown to the nearest whole number



The relative atomic mass of lithium to two decimal places is 6.94 when rounded to the nearest whole number, the RAM is 7, which is the same as the mass number shown on this isotope of lithium

2.2 Ions & Ionic Bonds

2.2.1 Ions & Ionic Bonds

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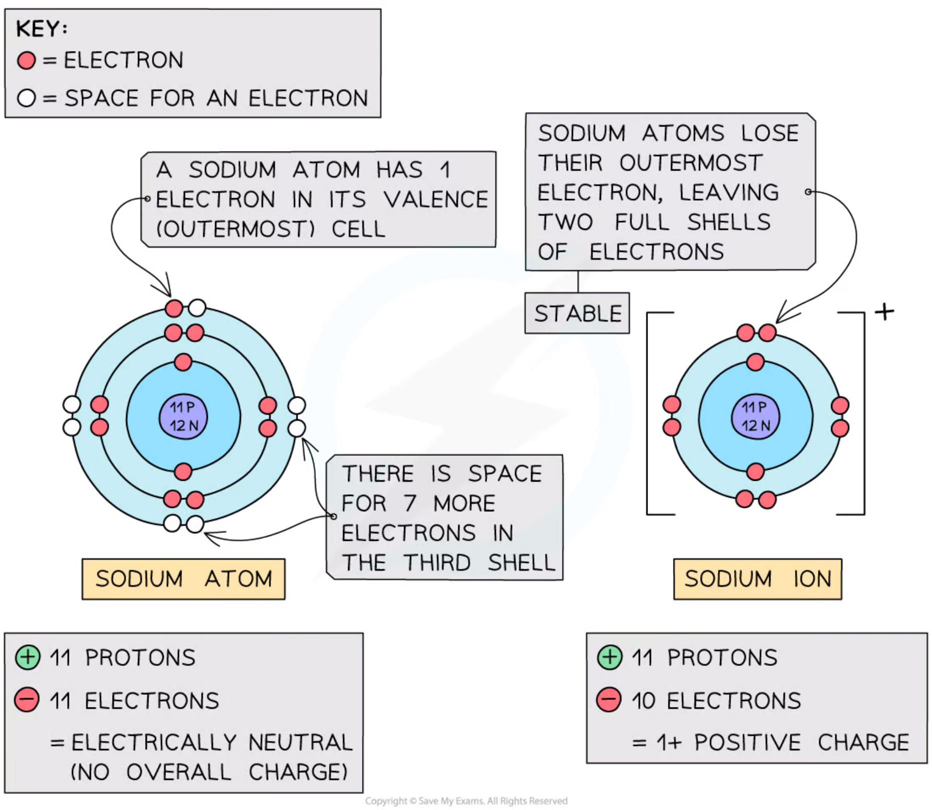
The Formation of Ions

- An ion is an **electrically charged** atom or group of atoms formed by the **loss** or **gain** of **electrons**
- An atom will lose or gain electrons to become more stable
- The loss or gain of electrons takes place to gain a **full outer shell** of electrons which is a more stable arrangement of electrons
- The electronic configuration of an ion will be the same as that of a noble gas – such as helium, neon and argon



Formation of positively charged sodium ion

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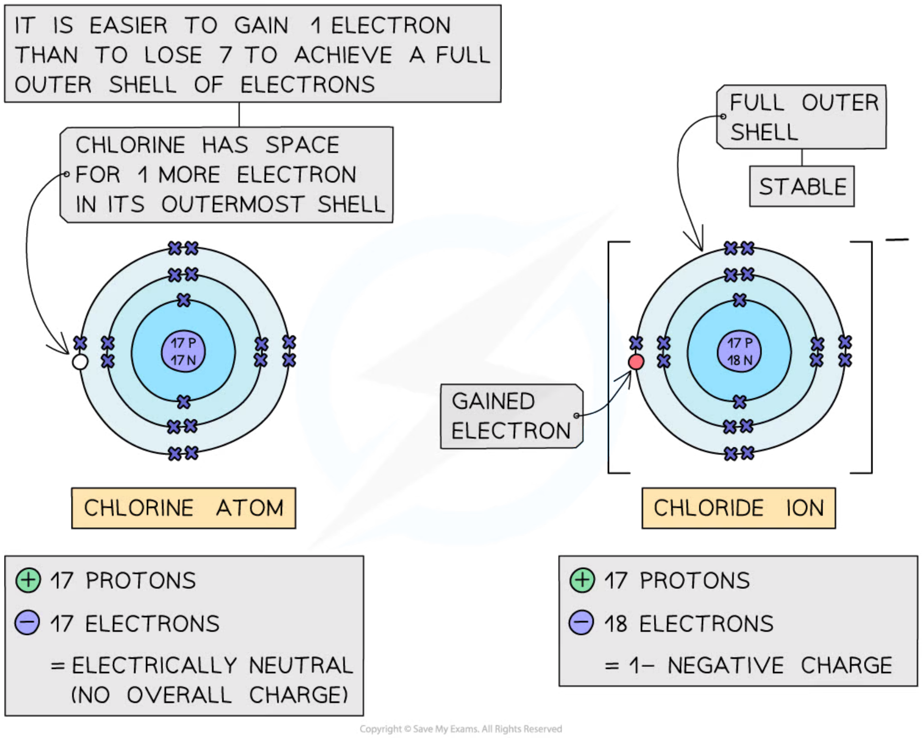

Formation of negatively charged chloride ion

Ionisation of metals and non-metals

- **Metals:** all metals can **lose** electrons to other atoms to become positively charged ions, known as **cations**
- **Non-metals:** all non-metals can **gain** electrons from other atoms to become negatively charged ions, known as **anions**

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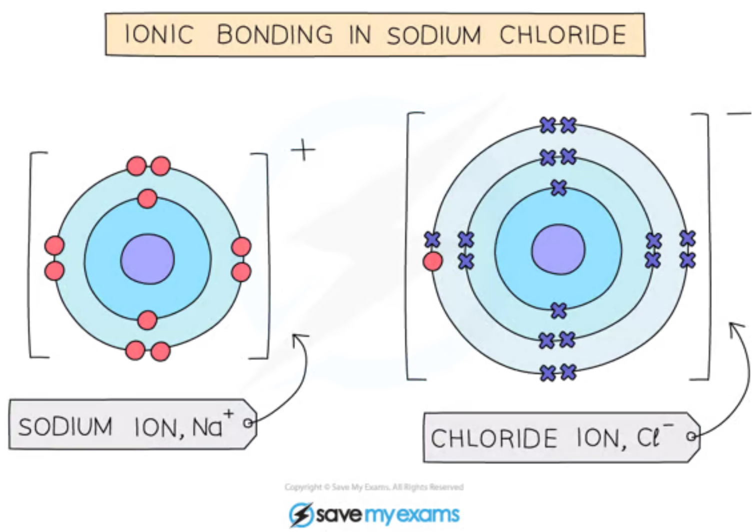
The Formation of Ionic Bonds

- Ionic compounds are formed when metal atoms react with non-metal atoms
- Metal atoms **lose** their outer electrons which the non-metal atoms **gain** to form positive and negative ions
- The positive and negative ions are held together by strong **electrostatic forces of attraction** between **opposite** charges
- This force of attraction is known as an **ionic bond** and they hold ionic compounds together

Dot-and-cross diagrams

- **Dot and cross diagrams** are diagrams that show the arrangement of the outer-shell electrons in an **ionic** or **covalent** compound or element
  - The electrons are shown as dots and crosses
- In a dot and cross diagram:
  - Only the outer electrons are shown
  - The charge of the ion is spread evenly which is shown by using brackets
  - The charge on each ion is written at the top right-hand corner



Electrostatic forces between the positive Na ion and negative Cl ion

Ionic Bonds between Group I & Group VII Elements

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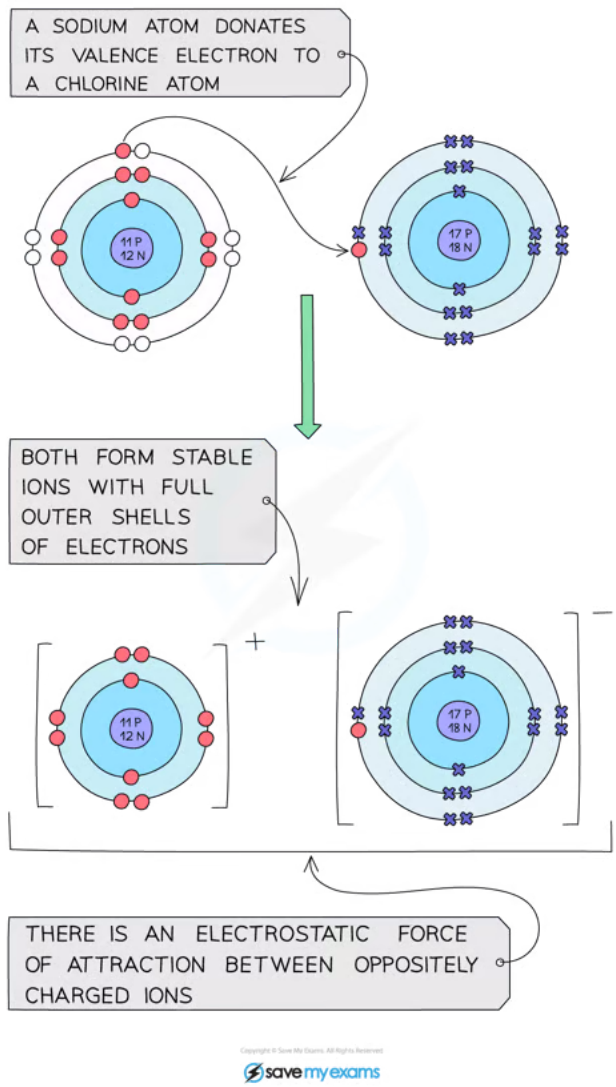
Example: Sodium Chloride, NaCl



Sodium chloride ionic bonding

Explanation

- Sodium is a Group I metal so will lose one outer electron to another atom to gain a full outer shell of electrons
- A positive sodium ion with the charge 1+ is formed
- Chlorine is a Group VII non-metal so will need to gain an electron to have a full outer shell of electrons
- One electron will be transferred from the outer shell of the sodium atom to the outer shell of the chlorine atom
- A chlorine atom will gain an electron to form a negatively charged chloride ion with a charge of 1–

- The oppositely charged ions are held together by strong electrostatic forces of attraction
- The ionic compound has no overall charge

Formula of ionic compound: $NaCl$

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2.2.2 Ionic Bonds & Lattice Structure

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The Lattice Structure of Ionic Compounds

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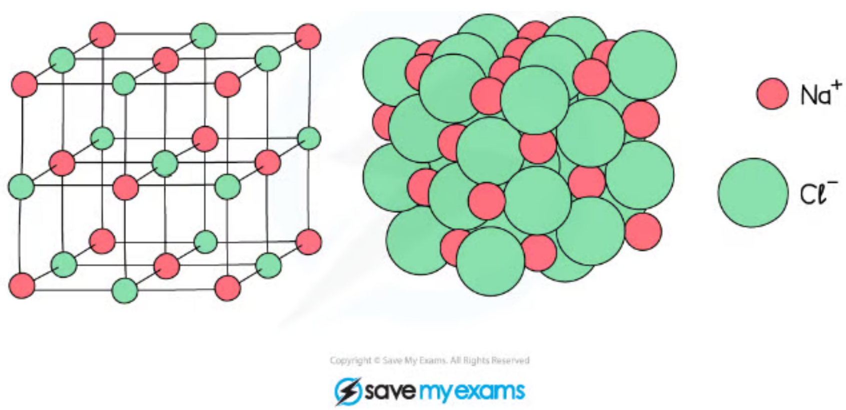
Lattice structure

- Ionic compounds have a **giant lattice** structure
- Lattice structure refers to the arrangement of the atoms of a substance in 3D space
- In lattice structures, the atoms are arranged in an **ordered** and **repeating** fashion
- The lattices formed by ionic compounds consist of a **regular arrangement** of **alternating** positive and negative ions



The lattice structure of NaCl

Ionic Bonds between Metallic & Non-Metallic Elements

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Ionic compounds

- Ionic compounds are formed when metal atoms and non-metal atoms react
- The ionic compound has **no** overall charge

Example: Magnesium Oxide, MgO

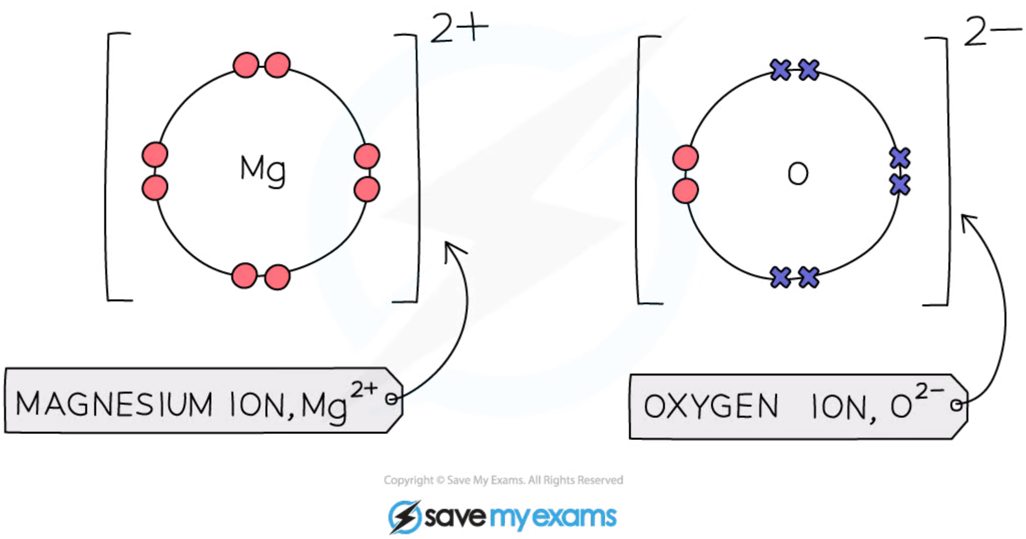

Diagram showing the dot-and-cross diagram of magnesium oxide

Explanation

- Magnesium is a Group II metal so will lose two outer electrons to another atom to have a full outer shell of electrons
- A positive ion with the charge 2+ is formed
- Oxygen is a Group VI non-metal so will need to gain two electrons to have a full outer shell of electrons
- Two electrons will be transferred from the outer shell of the magnesium atom to the outer shell of the oxygen atom
- Oxygen atom will gain two electrons to form a negative ion with charge 2-
- Magnesium oxide has no overall charge

Formula of ionic compound: MgO

Exam Tip

When drawing dot and cross diagrams, you only need to show the outer shell of electrons. Remember to draw square brackets and include a charge for each ion. Make sure the overall charge is 0; you may need to include more than one positive or negative ion to ensure the positive and negative charges cancel each other out.

2.2.3 Properties of Ionic Compounds

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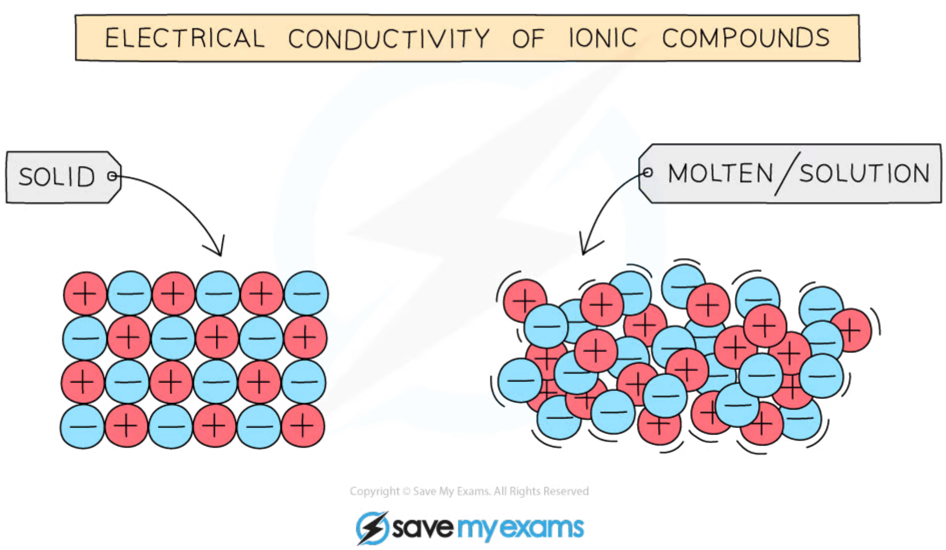
Properties of Ionic Compounds

- Ionic compounds are usually **solid** at room temperature
- They have **high** melting and boiling points
- Ionic compounds are good conductors of electricity in the **molten** state or in **solution**
- They are poor conductors in the solid state

Explaining the Properties of Ionic Compounds

EXTENDED

- Ionic substances have **high** melting and boiling points due to the presence of **strong electrostatic forces** acting between the oppositely charged ions
- These forces act in all directions and a lot of energy is required to overcome them
- The greater the charge on the ions, the stronger the electrostatic forces and the higher the melting point will be
  - For example, magnesium oxide consists of Mg^{2+} and O^{2-} so will have a higher melting point than sodium chloride which contains the ions, Na^{+} and Cl^{-}
- For electrical current to flow there must be freely moving charged particles such as electrons or ions present
- Ionic compounds are good conductors of electricity in the **molten** state or in **solution** as they have ions that can move and carry a charge
- They are poor conductors in the solid state as the ions are in fixed positions within the lattice and are unable to move



Molten or aqueous ions move freely but cannot in solid form

2.3 Simple Molecules & Covalent Bonds

2.3.1 Covalent Bonds

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The Formation of Covalent Bonds

Covalent compounds

- Covalent compounds are formed when **pairs** of electrons are **shared** between atoms
- Only **non-metal elements** participate in covalent bonding
- As in ionic bonding, each atom gains a **full outer shell** of electrons, giving them a noble gas electronic configuration
- When two or more atoms are **covalently** bonded together, we describe them as '**molecules**'
- **Dot-and-cross** diagrams can be used to show the electric configurations in simple molecules
- Electrons from one atom are represented by a dot, and the electrons of the other atom are represented by a cross
- The electron shells of each atom in the molecule overlap and the shared electrons are shown in the area of overlap
- The dot-and-cross diagram of the molecule shows clearly which atom each electron originated from

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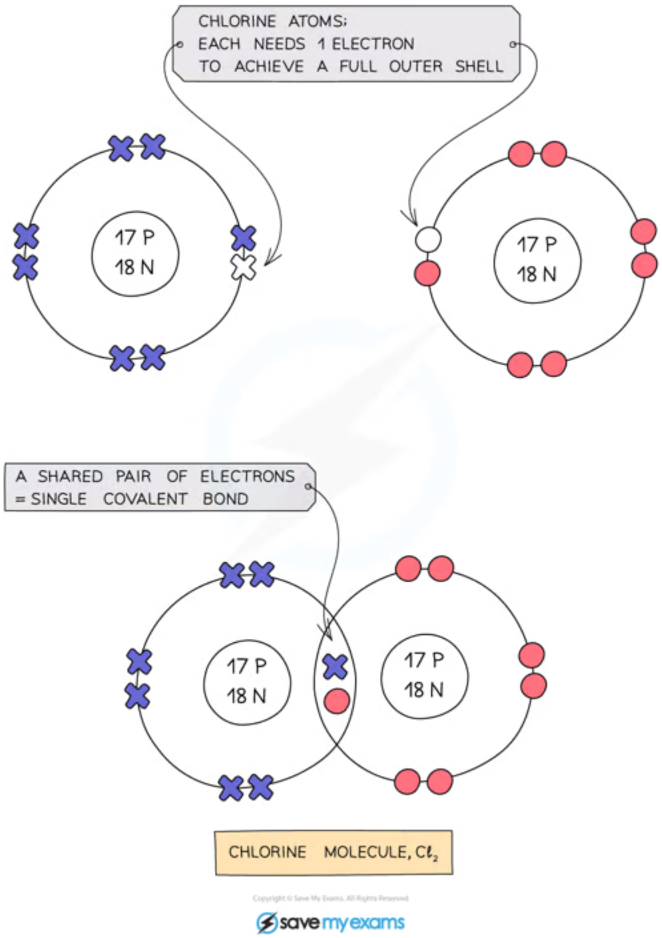


Diagram showing how a covalent bond forms between two chlorine atoms

Exam Tip

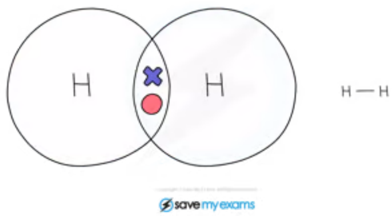
When drawing dot-and-cross diagrams for covalent compounds, make sure that the electron shell for each atom is full (remember that the 1st shell can only hold 2 electrons).

Single Covalent Bonds

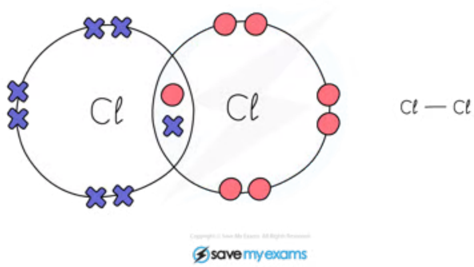
- Many simple molecules exist in which two adjacent atoms share one pair of electrons, also known as a single covalent bond (or single bond)

Common Examples of Simple Molecules

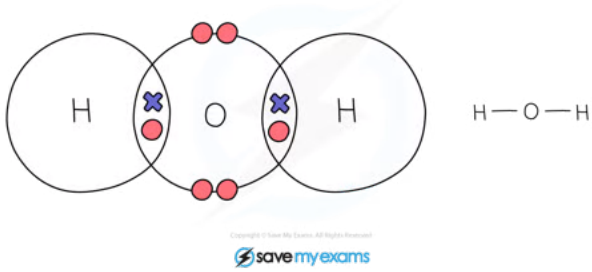
Hydrogen:



Chlorine:



Water:



Methane:

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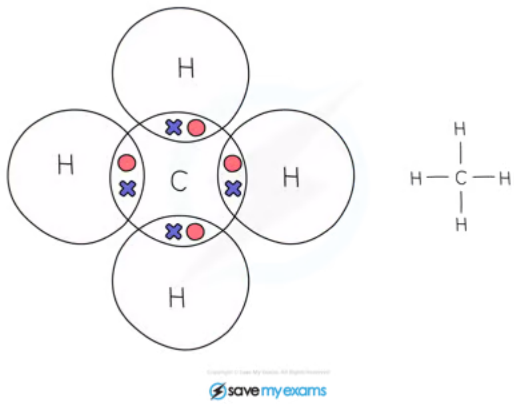
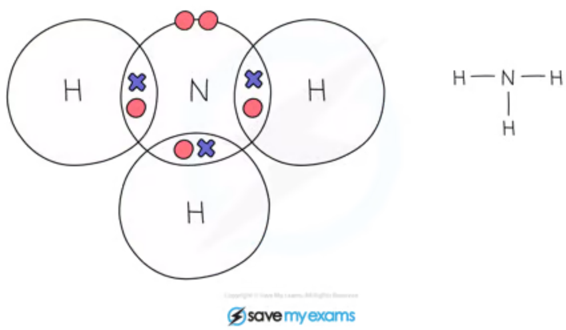
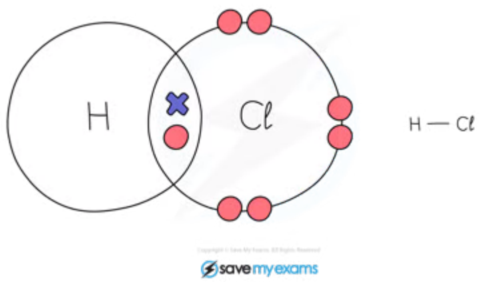


Ammonia:



Hydrogen chloride:



2.3.2 Molecules & Compounds

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Covalent Bonds in Complex Covalent Molecules

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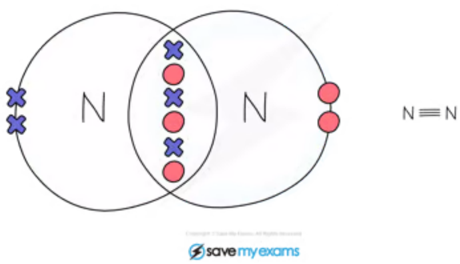
- Some atoms need to share more than one pair of electrons to gain a full outer shell of electrons
- If two adjacent atoms share two pairs of electrons, two covalent bonds are formed, also known as a **double bond**
- If two adjacent atoms share three pairs of electrons, three covalent bonds are formed, also known as a **triple bond**

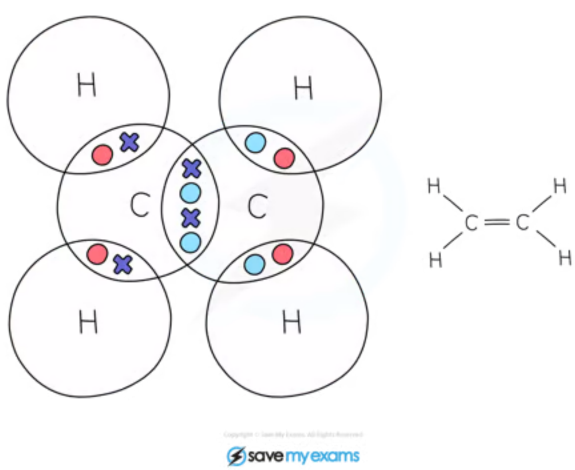
Nitrogen:

- When 2 nitrogen atoms react they share 3 pairs of electrons to form a triple bond



Ethene:

- In ethene, the 2 carbon atoms share 2 pairs of electrons
- This is known as a double bond



Methanol:

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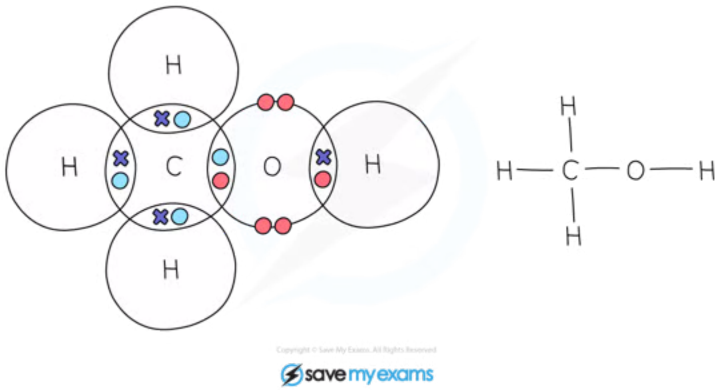
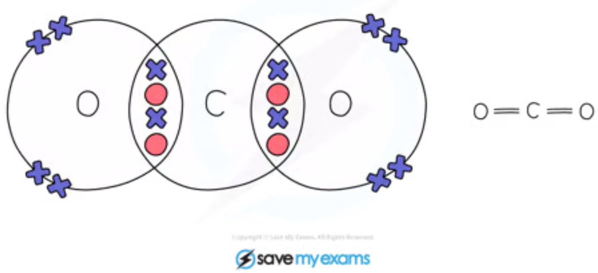


Carbon Dioxide:



Exam Tip

Be careful when drawing dot-and-cross diagrams, it is a common mistake for students to draw the wrong type of diagram. Remember, if the compound contains **metal** and **non-metal**, it is an **ionic** compound and you need to draw the ions separated, with square brackets around each ion, together with a charge. If the compound contains **non-metal** atoms only, it is a **covalent** compound, the shells should overlap and contain one or more pairs of electrons.

2.3.3 Properties of Simple Molecular Compounds

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Properties of Simple Molecular Compounds

- Small molecules are compounds made up of molecules that contain just a few atoms **covalently bonded** together
- They have **low** melting and boiling points so covalent compounds are usually **liquids** or **gases** at room temperature
- As the molecules increase in **size**, the melting and boiling points generally increase
- Small molecules have poor electrical conductivity

YOUR NOTES
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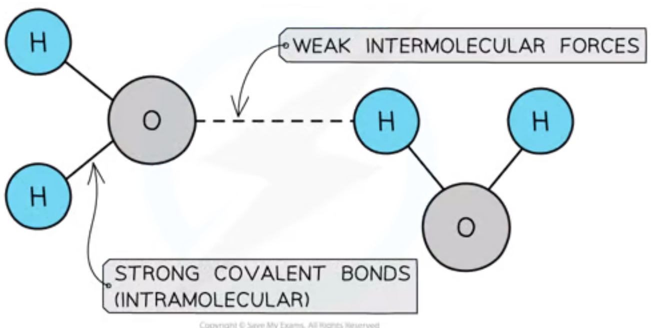
Explaining the Properties of Simple Molecular Compounds

EXTENDED

- Small molecules have covalent bonds joining the atoms together, but intermolecular forces that act between neighbouring molecules
- They have **low** melting and boiling points as there are only **weak intermolecular** forces acting **between** the molecules
- These forces are **very weak** when compared to the covalent bonds and so most small molecules are either gases or liquids at room temperature
- As the molecules increase in **size** the intermolecular forces also increase as there are more electrons available
- This causes the melting and boiling points to **increase**



The bonds between hydrogen and oxygen in water are COVALENT, and the attractions between the molecules are INTERMOLECULAR FORCES which are about one tenth as strong as covalent bonds

Exam Tip

The atoms within covalent molecules are held together by covalent bonds while the molecules in a covalent substance are attracted to each other by intermolecular forces.

Electrical Conductivity

Molecular compounds are **poor conductors** of electricity as there are no free ions or electrons to carry the charge.

- Most covalent compounds do not conduct at all in the solid state and are thus **insulators**
- Common insulators include the plastic coating around household electrical wiring, rubber and wood

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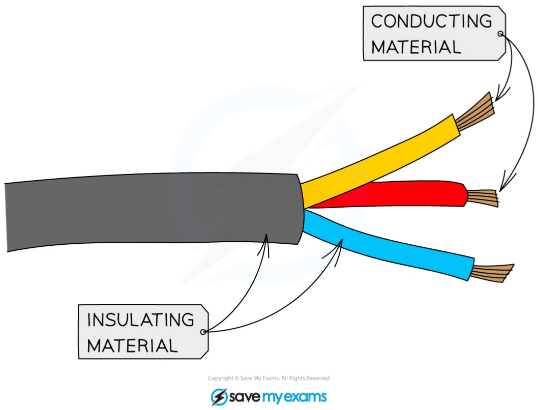

The plastic coating around electrical wires is made from covalent molecules that do not allow a flow of charge

YOUR NOTES
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2.4 Giant Structures

2.4.1 Diamond & Graphite

Structure of Graphite & Diamond

- Diamond and graphite are allotropes of carbon which have **giant covalent structures**
- Both substances contain only carbon atoms but due to the differences in bonding arrangements they are physically completely different
- Giant covalent structures contain billions of non-metal atoms, each joined to adjacent atoms by covalent bonds forming a giant lattice structure

Diamond

- In diamond, each carbon atom bonds with four other carbons, forming a **tetrahedron**
- All the covalent bonds are identical, very strong and there are no **intermolecular forces**

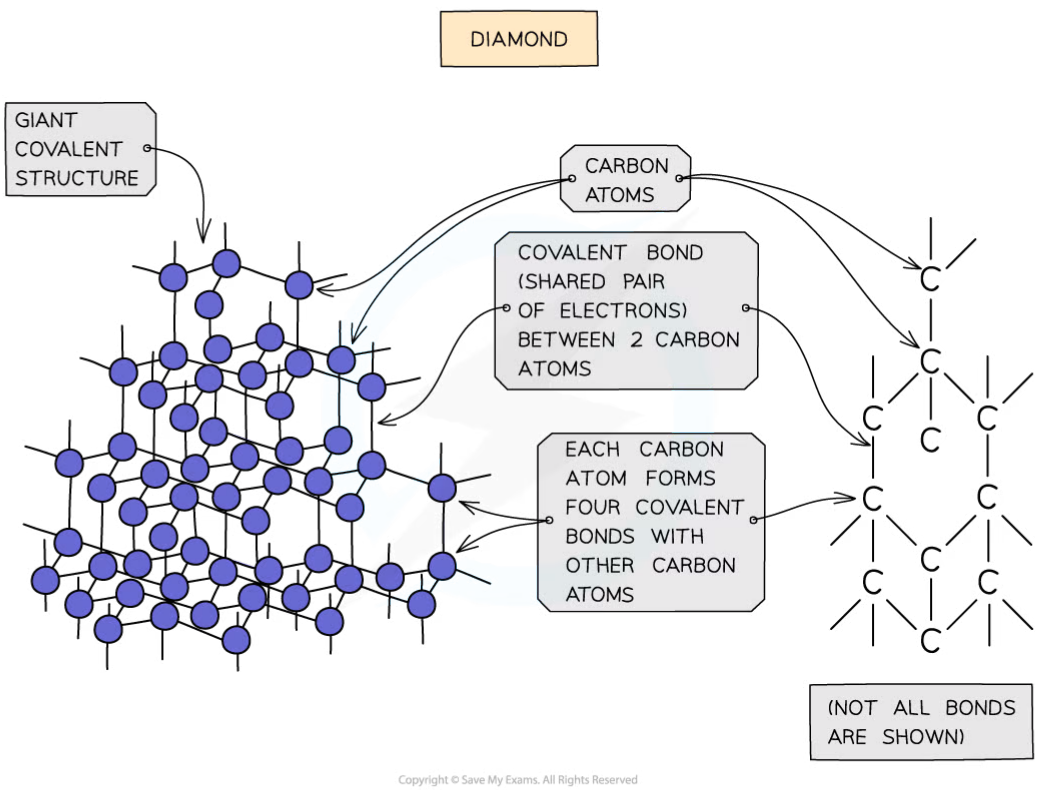
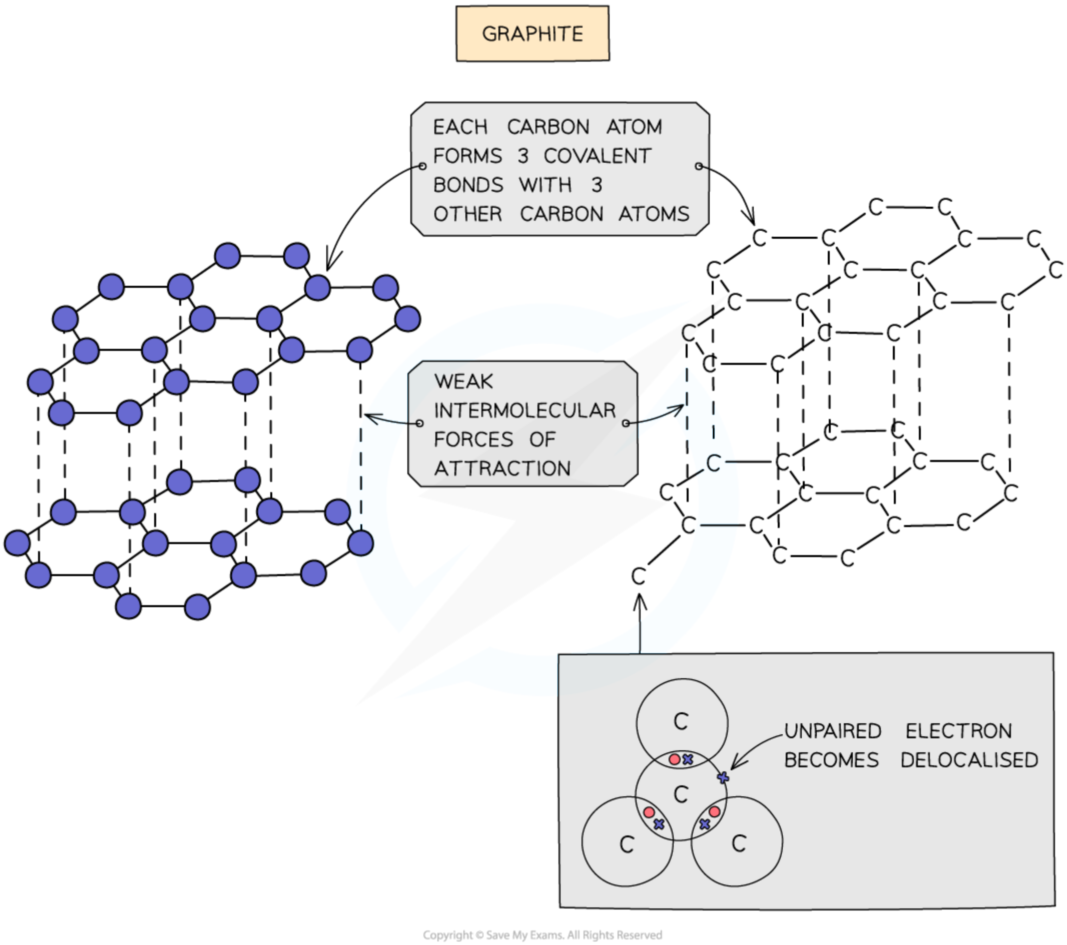


Diagram showing the structure and bonding arrangement in diamond

Graphite

- Each carbon atom in graphite is bonded to **three** others forming **layers** of **hexagons**, leaving one free electron per carbon atom which becomes delocalised

- The covalent bonds within the layers are very strong, but the layers are attracted to each other by weak **intermolecular forces**



The structure and bonding in graphite

YOUR NOTES
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Uses of Graphite & Diamond

YOUR NOTES
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Properties of Diamond

- Diamond has the following physical properties:
  - It does not conduct electricity
  - It has a very high melting point
  - It is extremely hard and dense

- All the outer shell electrons in carbon are held in the four covalent bonds around each carbon atom, so there are no freely moving charged particles to carry the current thus it cannot conduct electricity
- The four covalent bonds are very strong and extend in a giant lattice, so a very large amount of heat energy is needed to break the lattice thus it has a very high melting point
- Diamond´s hardness makes it very useful for purposes where extremely tough material is required
- Diamond is used in **jewellery** due to its sparkly appearance and as **cutting tools** as it is such a hard material
- The cutting edges of discs used to cut bricks and concrete are tipped with diamonds
- Heavy-duty drill bits and tooling equipment are also diamond-tipped

Exam Tip

Diamond is the hardest naturally occurring mineral, but it is by no means the strongest. Students often confuse **hard** with **strong**, thinking it is the opposites of weak. Diamonds are hard, but brittle - that is, they can be smashed fairly easily with a hammer. The opposite of saying a material is hard is to describe it as **soft**.

Properties of Graphite

- Each carbon atom is bonded to **three** others forming **layers** of hexagonal-shaped forms, leaving one free electron per carbon atom
- These free (delocalised) electrons exist **in between** the layers and are free to move through the structure and carry charge, hence graphite can conduct electricity
- The covalent bonds within the layers are very strong but the layers are connected to each other by **weak forces** only, hence the layers can **slide** over each other making graphite **slippery** and **smooth**
- Graphite thus:
  - Conducts electricity
  - Has a very high melting point
  - Is soft and slippery, less dense than diamond

- Graphite is used in **pencils** and as an industrial **lubricant**, in engines and in locks
- It is also used to make non-reactive **electrodes** for **electrolysis**

Exam Tip

Don't confuse pencil lead with the metal lead – they have nothing in common. Pencil lead is actually graphite, and historical research suggests that in the past, lead miners sometimes confused the mineral galena (lead sulfide) with graphite; since the two looked similar they termed both minerals 'lead'. The word graphite derives from the Latin word 'grapho' meaning 'I write', so it is a well named mineral!

YOUR NOTES
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2.4.2 Silicon(IV) Oxide

YOUR NOTES

Structure of Silicon(IV) Oxide

EXTENDED

- Silicon(IV) oxide (also known as silicon dioxide or silica), SiO_2, is a macromolecular compound which occurs naturally as **sand** and **quartz**
- Each oxygen atom forms covalent bonds with **2** silicon atoms and each silicon atom in turn forms covalent bonds with **4** oxygen atoms
- A tetrahedron is formed with one silicon atom and four oxygen atoms, similar to diamond

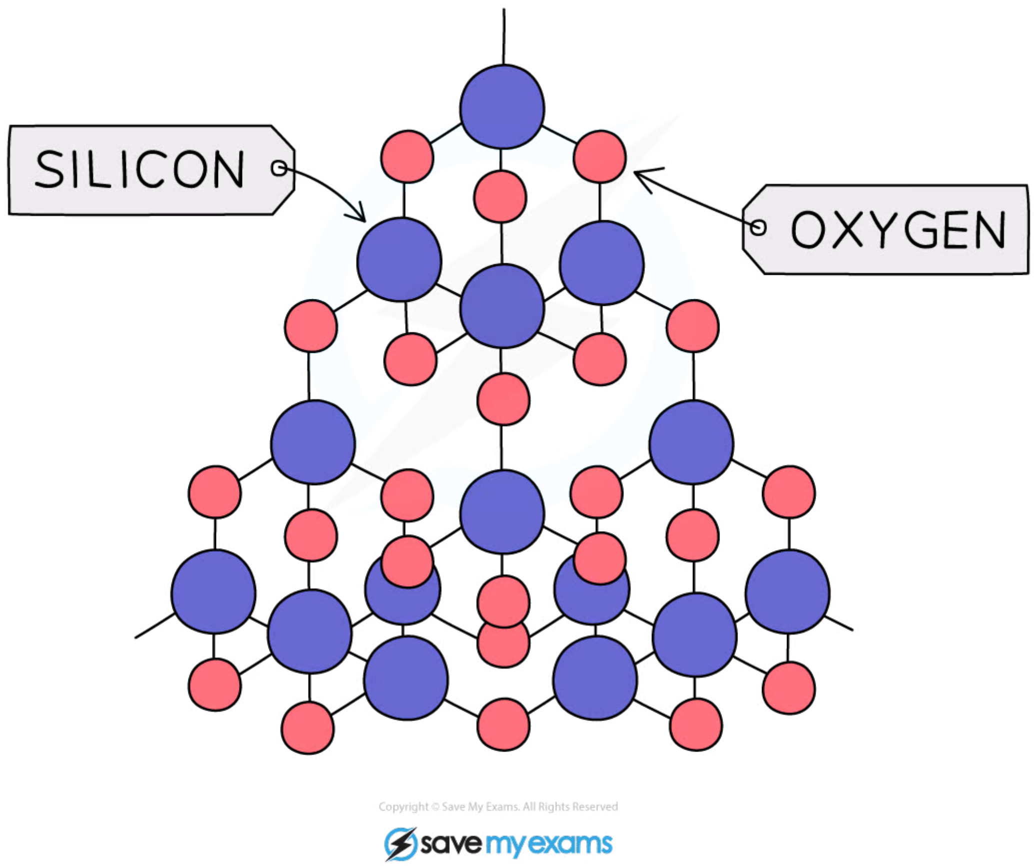

Diagram showing the structure of SiO_2 with the silicon atoms in blue and the oxygen atoms in red

Comparing Diamond & Silicon(IV) Oxide

YOUR NOTES
⬇

EXTENDED

- SiO_2 has lots of very strong covalent bonds and no intermolecular forces so it has similar properties to diamond
- It is very **hard**, has a very **high** boiling point, is insoluble in water and does not conduct electricity
- SiO_2 is cheap since it is available naturally and is used to make sandpaper and to line the inside of furnaces

2.4.3 Metallic Bonding

YOUR NOTES

Metallic Bonding

EXTENDED

- Metal atoms are held together strongly by metallic bonding in a giant metallic lattice
- Within the metallic lattice, the atoms lose the electrons from their outer shell and become positively charged ions
- The outer electrons no longer belong to a particular metal atom and are said to be **delocalised**
- They move freely between the positive metal ions like a 'sea of electrons'
- Metallic bonds are strong and are a result of the attraction between the positive metal ions and the negatively charged delocalised electrons

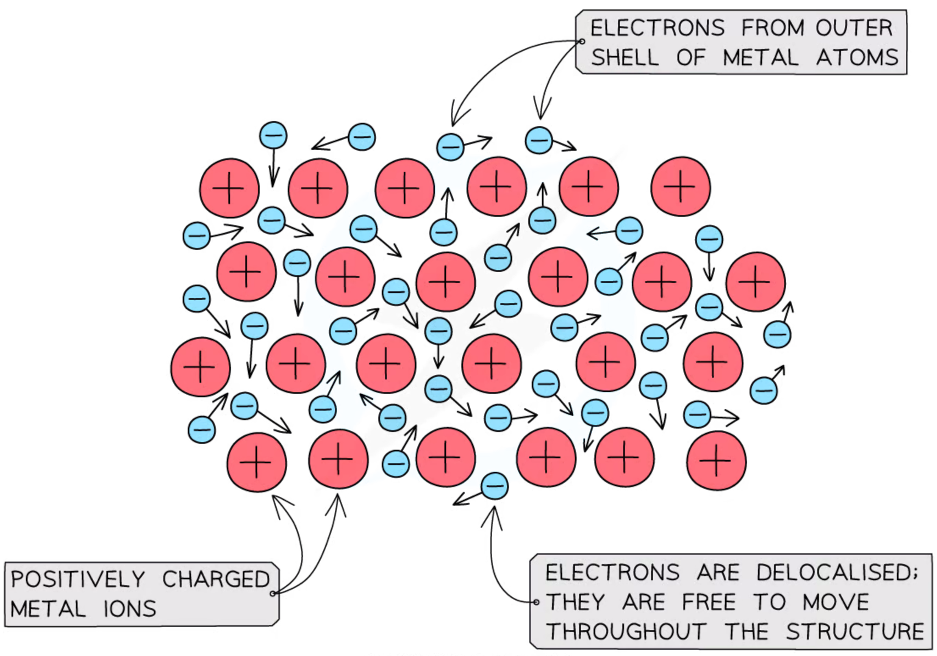

Diagram showing metallic lattice structure with delocalised electrons

Properties of Metals

YOUR NOTES

EXTENDED

- Metals have **high** melting and boiling points
  - There are many **strong metallic bonds** in giant metallic structures between the positive metal ion and delocalised electrons
  - A lot of heat energy is needed to break these bonds
- Metals **conduct** electricity
  - There are **free electrons** available to move through the structure and carry charge
  - Electrons entering one end of the metal cause a delocalised electron to displace itself from the other end
  - Hence electrons can flow so electricity is conducted
- Metals are **malleable** and **ductile**
  - Layers of positive ions can **slide** over one another and take up different positions
  - Metallic bonding is not disrupted as the outer electrons do not belong to any particular metal atom so the delocalised electrons will move with them
  - Metallic bonds are thus not broken and as a result metals are strong but **flexible**
  - They can be hammered and bent into different shapes or drawn into wires without breaking

Exam Tip

When explaining why metals can conduct electricity, be careful of the terminology you use. Don't get confused with ionic compounds. **Metals** can conduct electricity as they have free **electrons** that can carry charge whereas molten or aqueous **ionic** compounds can conduct electricity because they have free **ions** that can carry charge.